

Aditya Meenakshisundaram

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Education

Texas A&M University

Expected May 2028

Bachelor of Science in Chemical Engineering

Cumulative GPA: 3.91 / 4.00 | Craig & Galen Brown Engineering Honors

Work Experience & Research

Photolithography & Metrology Intern

May 2026 – August 2026

Samsung Semiconductor

Austin, TX

- Engineered a Python-based overlay-correction tool that converted 500+ wafer- and shot-level measurements per lot into scanner corrections, resolving multi-layer misalignments and saving 250 engineering hours annually
- Deployed a PyTorch CNN trained on FFT-derived heatmaps to classify spinner-tool defects, automating real-time review of 1,000+ images daily
- Restored throughput by 300–400 wafers/day using I-MR control charts to isolate a downstream track bottleneck and resolve a degraded buffer-unit transfer mechanism
- Automated resets of lot-level overlay metrology skip factors to process-type defaults using Python and SQL, eliminating manual reverts and saving 50 department-hours per week

Samsung Semiconductor Research Fellow

Jan 2026 – Present

Samsung Semiconductor & Artie McFerrin Department of Chemical Engineering

College Station, TX

- Project: lithium-ion battery thermal-runaway mitigation using a sprayable, passive thermal-switching composite. Designed to switch from heat conduction to insulation via indium melt-out, with CaCO_3 releasing CO_2 for fire mitigation & a retained titanium framework for structural support
- Executed synthesis end to end: mixed CaCO_3 , indium, and titanium nanowires, compacted pellets, and heat-treated at 900 °C for 1 hour; examined structure and conductive pathways using SEM cross-sections across 4+ prototype iterations
- Compared heat-transfer response across 50–70 wt% indium pellets using hot-plate heating and top-surface temperature measurements; identified 65 wt% as the best-performing formulation tested, with improved structural integrity

Silicon Carbide (SiC) Researcher

Aug 2025 – Present

Artie McFerrin Department of Chemical Engineering, Texas A&M University

College Station, TX

- Project: conversion of liquid polycarbosilane (PCS) into electrically conductive silicon carbide for RF susceptor applications; targeted the lowest pyrolysis temperature needed to obtain conductivity
- Executed synthesis end to end, from liquid PCS through degassing, curing, resting, and pyrolysis; completed 10+ controlled runs at 800–1200 °C, varying temperature, ramp rate, and atmosphere
- Narrowed the onset of measurable electrical conductivity to pyrolysis temperatures between 900 and 1000 °C using four-point probe measurements, guiding subsequent trials within that range
- Characterized ceramic yield and microstructure using TGA and SEM; evaluated RF heating response to assess how conversion temperature affected electromagnetic coupling and heat generation

Projects

STEM Research Finder

Co-Creator; Lead Developer

stemresearchfinder.tech

- Co-created and launched a multi-tenant research-discovery platform supporting university-specific deployments; indexed 5,000+ STEM faculty nationwide and attracted 3,000+ student users without paid acquisition
- Built Python data pipelines across 50+ university web sources to extract and standardize faculty research profiles, deduplicate records, and automate index updates
- Developed resume-based faculty matching and personalized outreach drafting, connecting student experience with relevant research labs and streamlining advisor outreach

Substack Author

adimeenak.substack.com

- Published 30+ articles on semiconductors, materials & energy; 400+ monthly readers

Honors & Technical Skills

Honors & Awards: Humba Deep Tech Fellowship (\$6,000 award), Samsung Semiconductor Research Fellowship (\$10,000 award), Samsung Semiconductor Intern Scholarship (\$10,000 award), TEX-E Fellowship, Omega Chi Epsilon

Characterization: Scanning Electron Microscopy, X-Ray Diffraction, Thermogravimetric Analysis, Optical Microscopy

Processing & Testing: Photolithography, Overlay Metrology, Pyrolysis, Thermal Testing, Statistical Process Control

Programming & Data: Python(NumPy, Pandas, SciPy, PyTorch, TensorFlow), SQL, MATLAB, Git, Docker, Linux